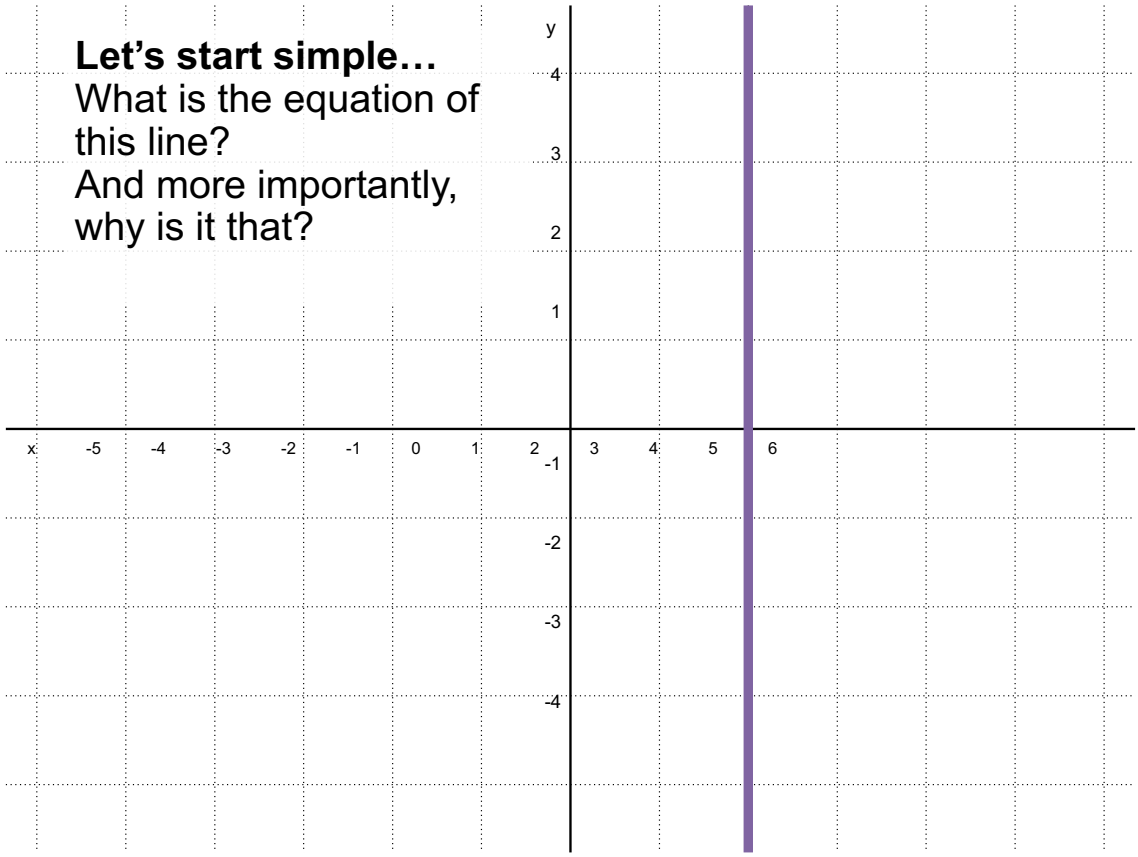
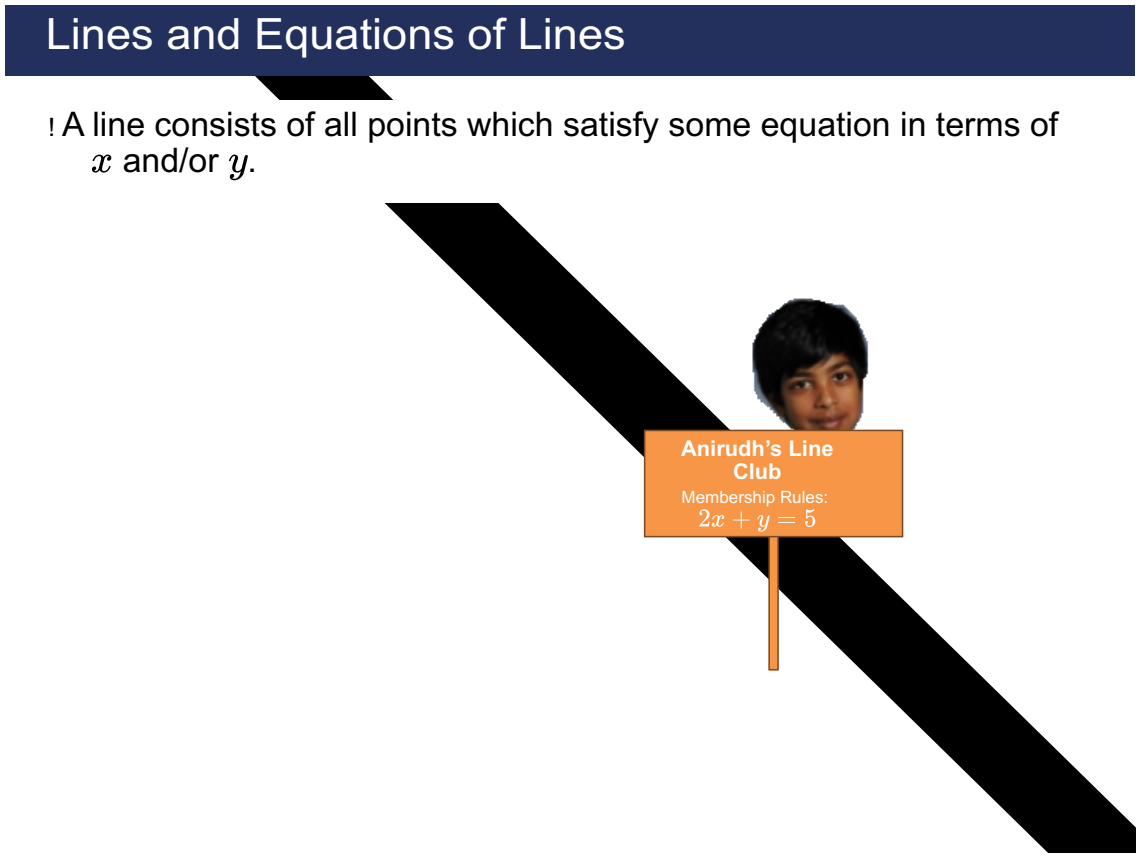
Coordinate Geometry

Chapter 5 · Original slide 4



Chapter 5 · Original slide 5



Coordinate Geometry

Chapter 5 · Original slide 6

Examples

This means we can **substitute** the values of a coordinate into our equation whenever we know the point lies on the line.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of a.

Find the coordinate of the point where the line $2x + y = 5$ cuts the x-axis.

Chapter 5 · Original slide 7

Test Your Understanding

Determine where the line $x + 2y = 3$ crosses the:

a) y-axis:

b) x-axis:

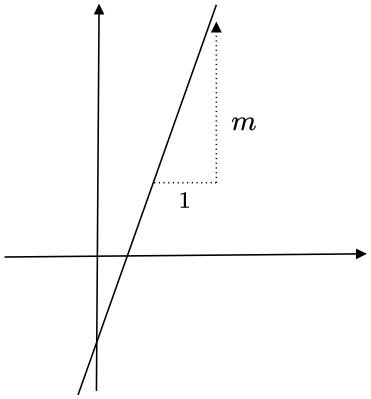
What mistakes do you think it's easy to make?

Coordinate Geometry

Chapter 5 · Original slide 8

Recap of gradient

The steepness of a line is known as the **gradient**.
It tells us what y changes by as x increases by 1.



So if the y value increased by 6 as the x value increased by 2, what is y increasing by for each unit increase of x?
How would that give us a suitable formula for the gradient m?

$$m = \frac{\Delta y}{\Delta x}$$

Chapter 5 · Original slide 9

Examples

Find the gradient of the line that goes through the points:

1. $(1, 4)$ $(3, 10)$
2. $(5, 7)$ $(8, 1)$
3. $(2, 2)$ $(-1, 10)$
4. Show that the points $A(3, 4), B(5, 5), C(11, 8)$ all lie on a straight line.

If points are 'collinear' they lie on the same line.

Coordinate Geometry

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Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1. Work out the value of a.

Chapter 5 · Original slide 11

$y = mx + c$

One form we can put a straight line equation in is:

$$y = mx + c$$



Why does it work?

- **The $\mathbf{y}$-intercept by definition is the $\mathbf{y}$ value when $\mathbf{x=0}$.**
  Substituting:
  $\mathbf{y=(m\times 0)+c=c}$ **as expected.**
- **By the definition of gradient, if we increase $\mathbf{x}$ by 1, $\mathbf{y}$ should increase by $\mathbf{m}$:**
  $\mathbf{y=m\,(x+1)+c=mx+m+c}$
  which indeed has increased by $\mathbf{m}$.

Coordinate Geometry

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$y = mx + c$

Determine the gradient and y-intercept of the line with equation $4x - 3y + 5 = 0$

Make y the subject so we have the form $y = mx + c$
Put y on the side it's positive.

Chapter 5 · Original slide 13

$ax + by + c = 0$

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the '**slope-intercept form**'.
But another common form is $\mathbf{ax+by+c=0}$, where a, b, c are integers.
This is known as the '**standard**' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where a, b, c are integers.

We don't want fractions, so multiply by an appropriate number.

Put everything on either side of equation.
$-x + 3y + 2 = 0$ would also be OK.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$...

Coordinate Geometry

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Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$y = mx + c$
"Slope-Intercept Form"

$ax + by + c = 0$
"Standard Form"

Coverage

$y = mx + c$ doesn't allow you to represent vertical lines. Standard form allows us to do this by just making b zero.

$x + 4 = 0$

Symmetry

In general, the '**linear combination**' of two variables x and y is $ax + by$, i.e. "some amount of x and some amount of y". There is a greater elegance and symmetry to this form over $y = mx + c$ because x and y appear similarly within the expression.

Usefulness

This more 'elegant' form also means it ties in with vectors and matrices. In FM, you will learn about the '**dot product**' of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

thus since $ax + by + c = 0$, we can represent a straight line using:

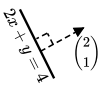
$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 \qquad (1)$$

We can extend to 3D points to get the equation of a **plane**:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 \qquad (2)$$

Conveniently, in equation (1), the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is **perpendicular to the line**. And in equation (2), the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$2x + y = 4$ ➡ $\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$ ➡



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Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where a, b, c are integers.

Coordinate Geometry

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Exercise 5A – Pure Y1 Page 91

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. $(3\ \mathrm{ma}\ldots$

Chapter 5 · Original slide 17

Exercise 5B – Pure Y1 Page 92-93

8 A line l cuts the x-axis at $(5, 0)$ and the y-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the y-axis at $-\frac{c}{b}$.

10 The line l with gradient 3 and y-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

Find the values of a and c. (2 marks)

Coordinate Geometry

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Exercise 5B – Pure Y1 Page 92-93

11 The straight line l passes through $(0, 6)$ and has gradient -2. It intersects the line with equation $5x - 8y - 15 = 0$ at point P. Find the coordinates of P.

12 The straight line l_1 with equation $y = 3x - 7$ intersects the straight line l_2 with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of b.

b Find the value of a.

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Equations using two points/point + gradient

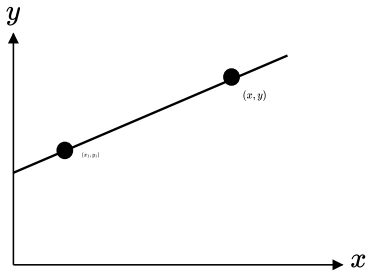
Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

How would you have done this at **GCSE**?

Coordinate Geometry

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A new way…



Notes: Note that x_1 and y_1 are constants while x and y are variables. The latter are variables because as these 'vary', we get different points on the line.

Suppose that (x_1, y_1) is some fixed point on the line that we specify (e.g. $(3, 5)$).
Suppose that (x, y) represents a generic point on the line, which is allowed to change as we consider different points on this line.
Then:

$m = \frac{y-y_1}{x-x_1}$

Thus:

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Quickfire Questions

In a nutshell: You can use this formula whenever you have (a) a gradient and (b) any point on the line.

Gradient	Point	(Unsimplified) Equation
3	$(1, 2)$	
5	$(3, 0)$	
2	$(-3, 4)$	
$\frac{1}{2}$	$(1, -5)$	
9	$(-4, -4)$	

Coordinate Geometry

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Using 2 points

Find the equation of the line that goes through $(4, 5)$ and $(6, 2)$, giving your equation in the form $ax + by + c = 0$.

We need to work out the gradient first before we can use $y - y_1 = m(x - x_1)$:

Test Your Understanding:
Find the equation of the line that goes through $(-1, 9)$ and $(4, 5)$, giving your equation in the form $ax + by + c = 0$.

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Exercise 5C – Pure Y1 Page 94-95

3 Find the equation of the line l which passes through the points $A(7, 2)$ and $B(9, -8)$. Give your answer in the form $ax + by + c = 0$.

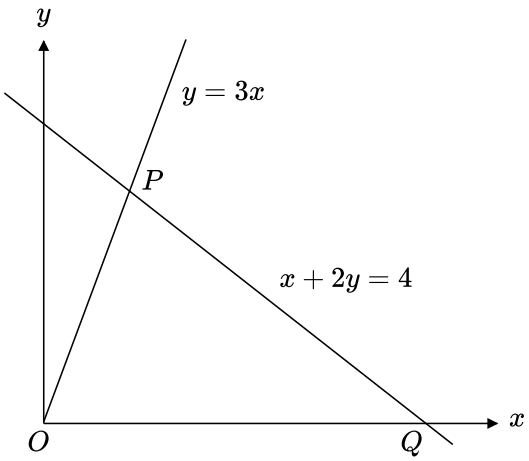
5 The straight line l passes through $(a, 4)$ and $(3a, 3)$. An equation of l is $x + 6y + c = 0$. Find the value of a and the value of c.

6 The straight line l passes through $(7a, 5)$ and $(3a, 3)$. An equation of l is $x + by - 12 = 0$. Find the value of a and the value of b.

Coordinate Geometry

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Intersection of lines



The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

a) Determine the coordinates of P.

We know from Chapter 4 we can just solve two equations simultaneously.

b) The line $x + 2y = 4$ intersects the x-axis at the point Q. Determine the coordinate of Q.

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Test Your Understanding

C1 Edexcel May 2013 Q6

The straight line L_1 passes through the points $(-1, 3)$ and $(11, 12)$.
(a) Find an equation for L_1 in the form $ax + by + c = 0$, where a, b and c are integers.
\hfill(4)
The line L_2 has equation $3y + 4x - 30 = 0$.
(b) Find the coordinates of the point of intersection of L_1 and L_2. \hfill(3)

(a)

(b)

Coordinate Geometry

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Exercise 5D – Pure Y1 Page 96

6 The line that passes through the points $(-3, -5)$ and $(4, 9)$ meets the y-axis at the point G. Work out the coordinates of the point G.

9 The lines $y = 4x - 10$ and $y = x - 1$ intersect at the point T. Find the equation of the line with gradient $-\frac{2}{3}$ that passes through the point T. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

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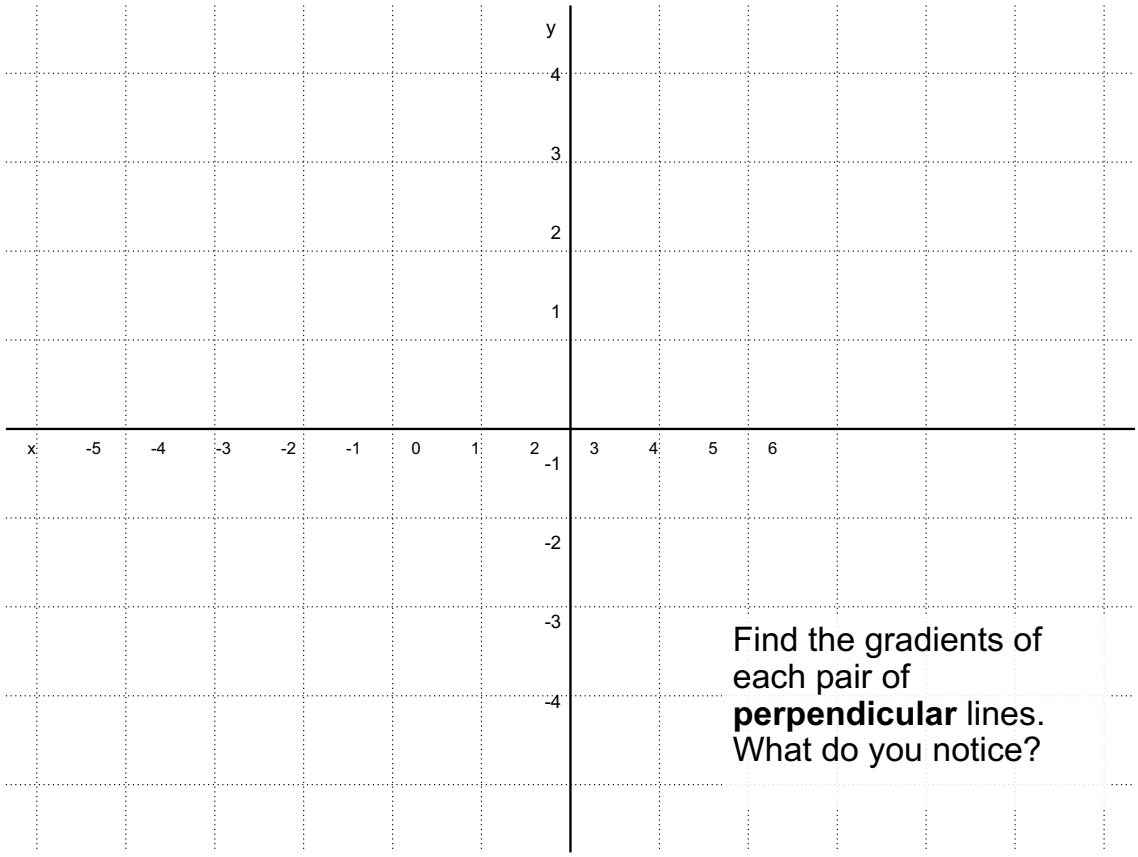
Exercise 5D – Pure Y1 Page 96

14 The line $y = 4x + 5$ meets the y-axis at the point C. The line $y = -3x - 15$ meets the x-axis at the point D. Find the equation of the line joining the points C and D. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

16 The lines $y = -2x + 1$ and $y = x + 7$ intersect at the point L. The point M has coordinates $(-3, 1)$. Find the equation of the line that passes through the points L and M.

Coordinate Geometry

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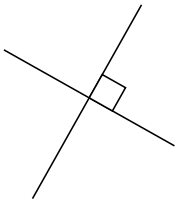
Perpendicular Lines

The gradients of parallel lines are equal.
If two lines are perpendicular, then the gradient of one is the **negative reciprocal** of the other.

$$m_1 = -\frac{1}{m_2}$$

To **show** that two lines are perpendicular:

$$m_1 m_2 = -1$$



Gradient	Gradient of Perpendicular Line
2	
-3	
$\frac{1}{4}$	
5	
$-\frac{2}{7}$	
$\frac{7}{5}$	

Coordinate Geometry

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Example Problems

1. A line is goes through the point (9,10) and is perpendicular to another line with equation $y = 3x + 2$. What is the equation of the line?

2. A line L_1 goes through the points $A(1, 3)$ and $B(3, -1)$. A second line L_2 is perpendicular to L_1 and passes through point B. Where does L_2 cross the x-axis?

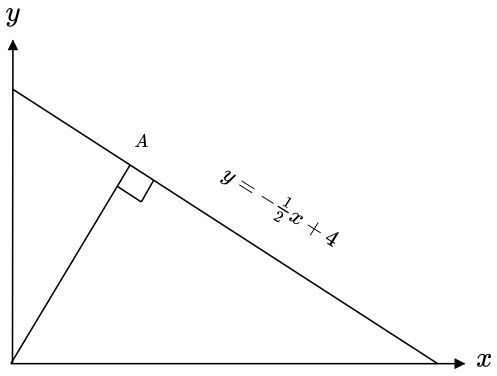
3. Are the following lines parallel, perpendicular, or neither?
$y = \frac{1}{2}x$
$2x - y + 4 = 0$

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Test Your Understanding

1. A line goes through the point (4,7) and is perpendicular to another line with equation y=2x+2. What is the equation of the line? Put your answer in the form ax+by+c=0, where a,b,c are integers.

2. Determine the point A.



Coordinate Geometry

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Exercise 5E/5F – Pure Y1 Page 96

3 The coordinates of a quadrilateral $ABCD$ are $A(-6, 2)$, $B(4, 8)$, $C(6, 1)$ and $D(-9, -8)$. Show that the quadrilateral is a trapezium.

7 A line is parallel to the line $2x - 3y - 1 = 0$ and it passes through the point $(0, 0)$. Write down the equation of the line.

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Exercise 5F – Pure Y1 Page 100

9 The line l passes through the points $(-3, 0)$ and $(3, -2)$ and the line n passes through the points $(1, 8)$ and $(-1, 2)$. Show that the lines l and n are perpendicular.

10 The vertices of a quadrilateral $ABCD$ have coordinates $A(-1, 5)$, $B(7, 1)$, $C(5, -3)$ and $D(-3, 1)$. Show that the quadrilateral is a rectangle.

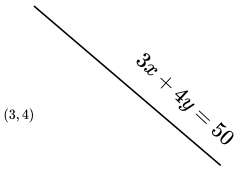
Coordinate Geometry

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Exercise 5E/5F – Pure Y1 Page 96

Extension Problems

- [MAT 2004 1D]
  What is the reflection of the point $(3, 4)$ in the line $3x + 4y = 50$?

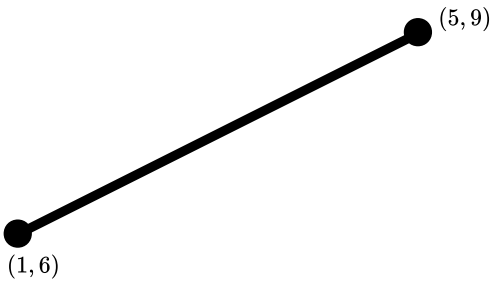


- [MAT 2014 1D] The reflection of the point $(1, 0)$ in the line $y = mx$ has coordinates: (in terms of m)

- *[STEP I 2004 Q6]* The three points A, B, C have coordinates $(p_1, q_1), (p_2, q_2)$ and (p_3, q_3), respectively. Find the point of intersection of the line joining A to the midpoint of BC, and the line joining B to the midpoint of AC. Verify that this point lies on the line joining C to the midpoint of AB.
  The point H has coordinates $(p_1 + p_2 + p_3, q_1 + q_2 + q_3)$. Show that if the line AH intersects the line BC at right angles, then $p_2^2 + q_2^2 = p_3^2 + q_3^2$, and write down a similar result if the line BH intersects the line AC at right angles.
  Deduce that if AH is perpendicular to BC and also BH is perpendicular to AC, then CH is perpendicular to AB.

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Distances between points



How could we find the **distance** between these two points?

Hint: Pythagoras

Form a right-angled triangle using the change in x and change in y, then use Pythagoras.

Coordinate Geometry

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Examples

Distance between:

(3,4) and $(5, 7)$
(5,1) and $(6, -3)$
(0,-2) and $(-1, 3)$

Note: Unlike with gradient, we don't care if the difference is positive or negative (it's being squared to make it positive anyway!)

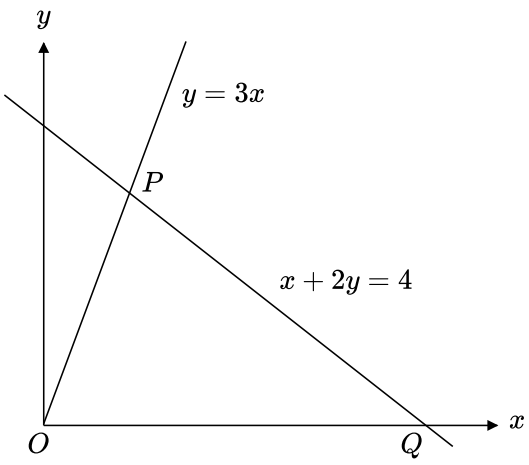
Quickfire Questions:

Distance between:

(1,10) and $(4, 14)$
(3,-1) and $(0, 1)$
(-4,-2) and $(-12, 4)$

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Area of Shapes



The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

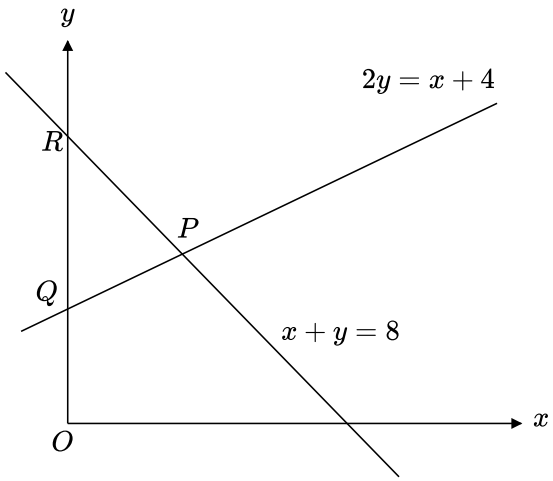
a) Determine the coordinates of P. (We did this in a previous lesson)

b) The line $x + 2y = 4$ intersects the x-axis at the point Q. Determine the area of the triangle OPQ.

Coordinate Geometry

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Further Example



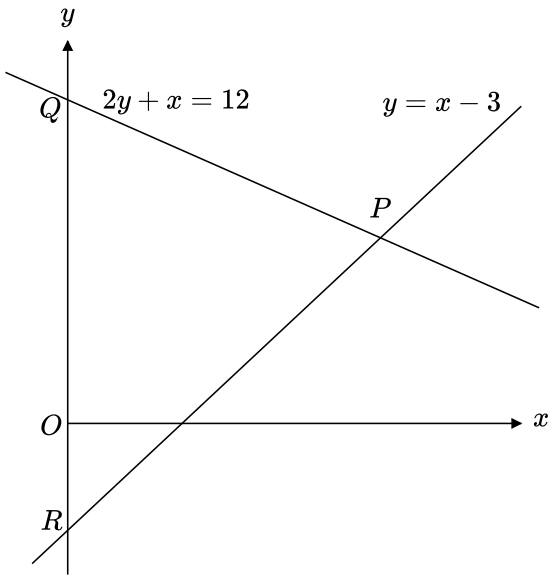
Tip: When finding areas of triangles in exam questions, one line is often vertical or horizontal. You should generally choose this to be the 'base' of your triangle.

a Determine the length of PQ.

b Determine the area PQR.

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Test Your Understanding



a) Determine the coordinate of P.

b) Determine the area of PQR.

c) Determine the length PQ.

Coordinate Geometry

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Exercise 5G – Pure Y1 Page 102

7 A point P lies on the line with equation $y = 4 - 3x$. The point P is a distance $\sqrt{34}$ from the origin. Find the two possible positions of point P. \hfill$(5\ \mathrm{ma}\ldots$

).

8 The vertices of a triangle are $A(2, 7)$, $B(5, -6)$ and $C(8, -6)$.
a Show that the triangle is a scalene triangle.
b Find the area of the triangle ABC.

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Exercise 5G – Pure Y1 Page 103

12 The straight line l_1 passes through the point $(-4, 14)$ and has gradient $-\frac{1}{4}$.
a Find an equation for l_1 in the form $ax + by + c = 0$, where a, b and c are integers.
b Write down the coordinates of A, the point where straight line l_1 crosses the y-axis.
The straight line l_2 passes through the origin and has gradient 3. The lines l_1 and l_2 intersect at the point B.
c Calculate the coordinates of B.
d Calculate the exact area of $\triangle OAB$.

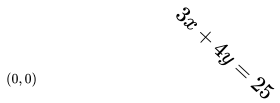
Coordinate Geometry

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Exercise 5G – Pure Y1 Page 102-103

Extension Problems

- [MAT 2001 1C]
  The shortest distance from the origin to the line $3x + 4y = 25$ is what?

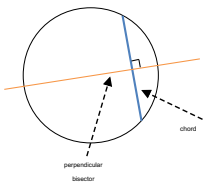


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Midpoints and Perpendicular Bisectors

4

Later in the chapter you will need to find the perpendicular bisector of a chord of a circle.



What two properties does a perpendicular bisector of two points A and B have?



Coordinate Geometry

Chapter 6 · Original slide 5

Test Your Understanding

5

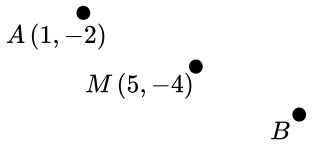
Find the perpendicular bisector of the line AB where A and B have the coordinates:
a) $A\,(4,7), B\,(10,17)$
b) $A\,(x_1,y_1), B\,(x_2,y_2)$

$M\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$

$m_{AB}=\frac{y_2-y_1}{x_2-x_1}\qquad m_{\perp}=\frac{x_1-x_2}{y_2-y_1}$

$y-\frac{y_1+y_2}{2}=\frac{x_1-x_2}{y_2-y_1}\left(x-\frac{x_1+x_2}{2}\right)$

A line segment AB is the diameter of a circle with centre $(5,-4)$. If A has coordinates $(1,-2)$, what are the coordinates of B?



We could use the formula $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$, but it is easy to see from the diagram that whatever movement there is from A to M, we can continue to get to B.

Chapter 6 · Original slide 6

Exercise 6A - Y1 Page 115

6

6 The line segment JK is a diameter of a circle, where J and K are $(\frac{3}{4},\frac{4}{3})$ and $(-\frac{1}{2},2)$ respectively. The centre of the circle lies on the line segment with equation $y=8x+b$. Find the value of b.

7 The line segment AB is a diameter of a circle, where A and B are $(0,-2)$ and $(6,-5)$ respectively. Show that the centre of the circle lies on the line $x-2y-10=0$.

Coordinate Geometry

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Exercise 6B - Y1 Page 116

7

2 The line FG is a diameter of the circle centre C, where F and G are $(-2, 5)$ and $(2, 9)$ respectively. The line l passes through C and is perpendicular to FG. Find the equation of l. **(7 marks)**

3 The line JK is a diameter of the circle centre P, where J and K are $(0, -3)$ and $(4, -5)$ respectively. The line l passes through P and is perpendicular to JK. Find the equation of l. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

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Exercise 6B - Y1 Page 117

8

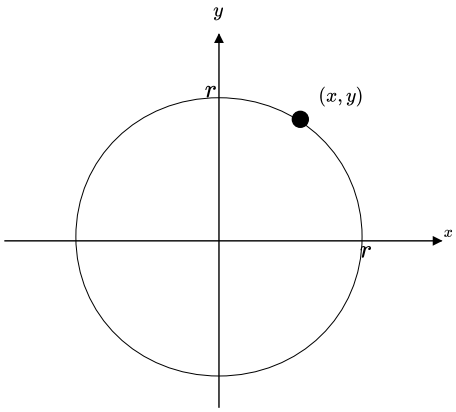
5 Point X has coordinates $(7, -2)$ and point Y has coordinates $(4, q)$. The perpendicular bisector of XY has equation $y = 4x + b$. Find the value of q and the value of b.

Coordinate Geometry

Chapter 6 · Original slide 9

Equation of a circle

9



Recall that a line can be a set of points (x, y) that satisfy some equation. Suppose we have a point (x, y) on a circle centred at the origin, with radius r. What equation must (x, y) satisfy?

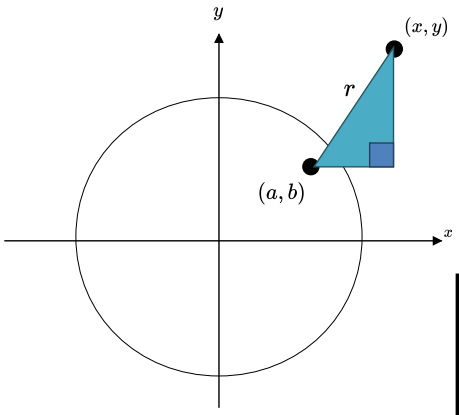
(Hint: draw a right-angled triangle inside your circle, with one vertex at the origin and another at the circumference)

$$x^2 + y^2 = r^2$$

Chapter 6 · Original slide 10

Equation of a circle

10



Now suppose we shift the circle so it's now centred at (a, b). What's the equation now?

(Hint: What would the sides of this right-angled triangle be now?)

The equation of a circle with centre (a, b) and radius r is:

$$(x - a)^2 + (y - b)^2 = r^2$$

Coordinate Geometry

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Quickfire Questions

11

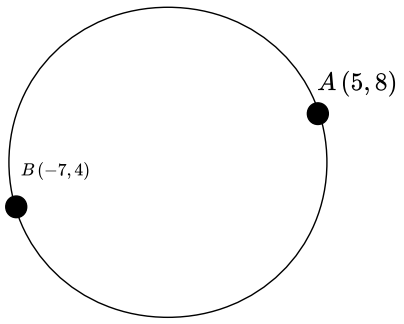
Centre	Radius	Equation
$(0, 0)$		

$(x+3)^2+(y-5)^2=1$
$(x+5)^2+(y-2)^2=49$
$(x+6)^2+y^2=16$
$(x-1)^2+(y+1)^2=3$
$(x+2)^2+(y-3)^2=8$

Chapter 6 · Original slide 12

Finding the equation using points

12



A line segment AB is the diameter of a circle, where A and B have coordinates $(5, 8)$ and $(-7, 4)$ respectively. Determine the equation of the circle.

Hint: What two things do we need to use the circle formula?

Coordinate Geometry

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Test Your Understanding

13

Edexcel C2 Jan 2005 Q2

The points A and B have coordinates $(5, -1)$ and $(13, 11)$ respectively.
(a) Find the coordinates of the mid-point of AB. \hfill(2)
Given that AB is a diameter of the circle C,
(b) find an equation for C. \hfill(4)

Chapter 6 · Original slide 14

Completing the square

14

When the equation of a circle is in the form $(x-a)^2 + (y-b)^2 = r^2$, we can instantly read off the centre (a, b) and the radius r.
But what if the equation wasn't in this form?

Find the centre and radius of the circle with equation $x^2 + y^2 - 6x + 2y - 6 = 0$

Hint: Have we seen a method in a previous chapter that allows us to turn a x^2 term and a x term into a single expression involving x?

Rearrange terms so that x terms are together and y terms are together.

Complete the square!

Coordinate Geometry

Chapter 6 · Original slide 15

Further Example

15

Edexcel C2 June 2012 Q3a,b

The circle C with centre T and radius r has equation
$x^2 + y^2 - 20x - 16y + 139 = 0$

(a) Find the coordinates of the centre of C. \hfill(3)
(b) Show that $r = 5$. \hfill(2)

Chapter 6 · Original slide 16

Example

16

Given the generic equation of a circle $x^2 + y^2 + ax + by + c = 0$ find a formula to determine the co-ordinate of the centre and the radius of the circle.

$$\left(x + \frac{a}{2}\right)^2 + \left(y + \frac{b}{2}\right)^2 = -c + \frac{a^2}{4} + \frac{b^2}{4}$$

$$r = \left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2 - c$$

$$c\left(-\frac{a}{2}, -\frac{b}{2}\right)$$

Coordinate Geometry

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Exercise 6C - Y1 Page 120

17

5 The line PQ is the diameter of the circle, where P and Q are $(5, 6)$ and $(-2, 2)$ respectively. Find the equation of the circle. \hfill(5 . . .

6 The point $(1, -3)$ lies on the circle $(x-3)^2 + (y+4)^2 = r^2$. Find the value of r.

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Exercise 6C - Y1 Page 120

18

8 a Show that $x^2 + y^2 - 4x - 11 = 0$ can be written in the form $(x-a)^2 + y^2 = r^2$, where a and r are numbers to be found. \hfill(2 marks)
b Hence write down the centre and radius of the circle with equation $x^2 +$
\hfill(2 marks)

9 a Show that $x^2 + y^2 - 10x + 4y - 20 = 0$ can be written in the form $(x-a)^2 + (y-b)^2 = r^2$, where a, b and r are numbers to be found. \hfill(2 m . . .
b Hence write down the centre and radius of the circle with equation $x^2 + y^2 - 10x + 4y - 20 =$
(2 m . . .

Coordinate Geometry

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Exercise 6C - Y1 Page 120

19

12 The point $P(7, -14)$ lies on the circle with equation $x^2 + y^2 + 6x - 14y = 17$. The point Q also lies on the circle such that PQ is a diameter. Find the coordinates of point Q.

13 The circle with equation $(x - k)^2 + y^2 = 41$ passes through the point $(3, 4)$. Find the two possible values of k.

Chapter 6 · Original slide 20

Extension

20

1 [MAT 2009 1B] The point on the circle $x^2 + y^2 + 6x + 8y = 75$ which is closest to the origin, is at what distance from the origin?

2 [MAT 2007 1D] The point on the circle $(x - 5)^2 + (y - 4)^2 = 4$ which is closest to the circle $(x - 1)^2 + (y - 1)^2 = 1$ has what coordinates?

3 [MAT 2016 1I] Let a and b be positive real numbers. If $x^2 + y^2 \leq 1$ then the largest that $ax + by$ can equal is what? Give your expression in terms of a and b.

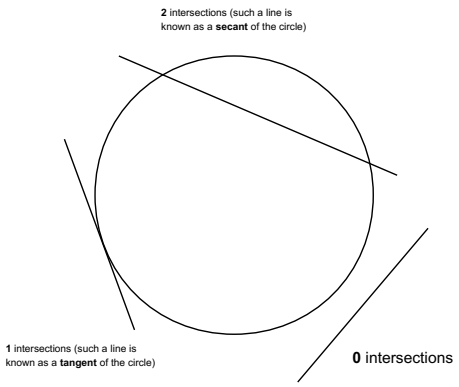
Coordinate Geometry

Chapter 6 · Original slide 21

Intersections of Lines and Circles

21

Recall that to consider the **intersection of two lines**, we attempt to solve them **simultaneously** by substitution, potentially using the **discriminant** to show that there are no solutions (and hence no points of intersection).



Show that the line $y = x + 3$ never intersects the circle with equation $x^2 + y^2 = 1$.

Chapter 6 · Original slide 22

Test Your Understanding

22

Find the points of intersection where the line $y = x + 6$ meets $x^2 + (y - 3)^2 = 29$.

Using an algebraic (and not geometric) method, determine the k such that the line $y = x + k$ **touches** the circle with equation $x^2 + y^2 = 1$.

Coordinate Geometry

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Exercise 6D - Y1 Page 122

23

6 a Show that the line $x + y = 11$ meets the circle with equation $x^2 + (y-3)^2 = 32$ at only one point. \hfill$(4\ \mathrm{mar}\ldots$

b Find the coordinates of the point of intersection. \hfill$(1\ \mathrm{ma}\ldots$

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Exercise 6D - Y1 Page 122

24

7 The line $y = 2x - 2$ meets the circle $(x-2)^2 + (y-2)^2 = 20$ at A and B.

a Find the coordinates of A and B.

b Show that AB is a diameter of the circle.

11 The line with equation $y = 4x - 1$ does not intersect the circle with equation $x^2 + 2x + y^2 = k$. Find the range of possible values of k.

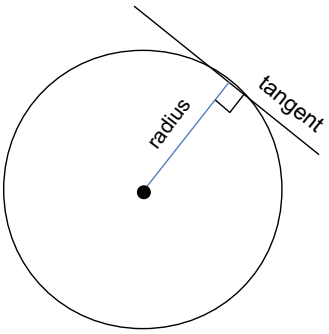
Coordinate Geometry

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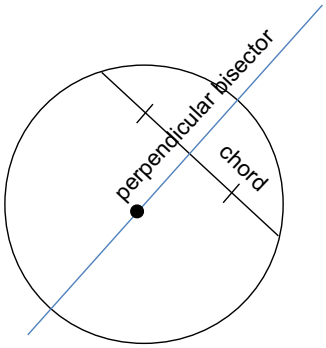
Tangents, Chords, Perpendicular Bisectors

25

There are two circle theorems that are of particular relevance to problems in this chapter, the latter you might be less familiar with:



The tangent is perpendicular to the radius (at the point of intersection).



The perpendicular bisector of any chord passes through the centre of the circle.

Why this will help:
If we knew the centre of the circle and the point of intersection, we can easily find the gradient of the radius, and thus the gradient and hence equation of the tangent.

Why this will help:
The first thing we did in this chapter is find the equation of the perpendicular bisector. If we had two chords, and hence found two bisectors, we could find the point of intersection, which would be the centre of the circle.

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Examples

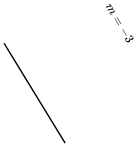
26

The circle C has equation $(x-3)^2+(y-7)^2=100$.
a) Verify the point $P\,(11,1)$ lies on C.
b) Find an equation of the tangent to C at the point P, giving your answer in the form $ax+by+c=0$

Tip: Use 'subscripting' of variables to make clear to the examiner (and yourself!) what you're calculating.

A circle C has equation $(x-4)^2+(y+4)^2=10$
The line l is a tangent to the circle and has gradient -3. Find two possible equations for l, giving your answers in the form $y=mx+c$.

This time we have the gradient, but don't have the points where the tangent(s) intersect the radius.



Coordinate Geometry

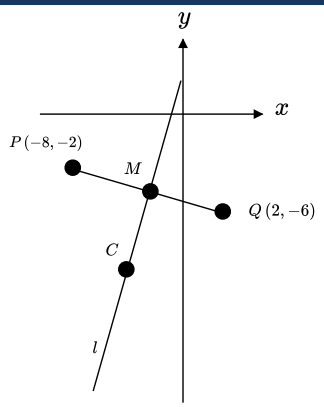
Chapter 6 · Original slide 27

Determining the Circle Centre

27

The points P and Q lie on a circle with centre C, as shown in the diagram. The point P has coordinates $(-8, -2)$ and the point Q has coordinates $(2, -6)$. M is the midpoint of the line segment PQ.
The line l passes through the points M and C.

a) Find an equation for l.
b) Given that the y-coordinate of C is -9:
i) show that the x-coordinate of C is -5.
ii) find an equation of the circle.



a We know the line going through the midpoint of a chord and a centre of the circle is the perpendicular bisector of the chord.

b

c

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Test Your Understanding

28

A circle has centre $C\,(3, 5)$, and goes through the point $P(6, 9)$. Find the equation of the tangent of the circle at the point P, giving your equation in the form $ax + by + c = 0$ where a, b, c are integers..

A circle passes through the points $A(0, 0)$ and $B\,(4, 2)$. The centre of the circle has x value -1. Determine the equation of the circle.

Coordinate Geometry

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Exercise 6E - Y1 Page 126-128

29

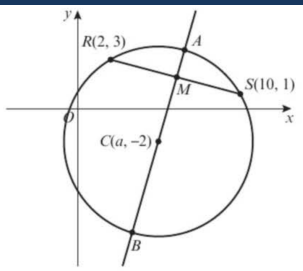
6 The tangent to the circle $(x+4)^2+(y-1)^2=242$ at $(7,-10)$ meets the y-axis at S and the x-axis at T.
a Find the coordinates of S and T. \hfill(5 marks)
b Hence, find the area of $\triangle OST$, where O is the origin. \hfill(3 marks)

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Exercise 6E - Y1 Page 126-128

30

10 The points R and S lie on a circle with centre $C(a,-2)$, as shown in the diagram.
The point R has coordinates $(2,3)$ and the point S has coordinates $(10,1)$.
M is the midpoint of the line segment RS.
The line l passes through M and C.
a Find an equation for l. (4 marks)
b Find the value of a. (2 marks)
c Find the equation of the circle. (3 marks)



d Find the points of intersection, A and B, of the line l and the circle. (5 marks)

Coordinate Geometry

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Exercise 6E - Y1 Page 126-128

31

Extension:

1 *[MAT 2012 1A]* Which of the following lines is a tangent to the circle with equation $x^2 + y^2 = 4$?
A) $x + y = 2$
B) $y = x - 2\sqrt{2}$
C) $x = \sqrt{2}$
D) $y = \sqrt{2} - x$

2 *[AEA 2006 Q4]* The line with equation $y = mx$ is a tangent to the circle C_1 with equation $(x + 4)^2 + (y - 7)^2 = 13$
(a) Show that m satisfies the equation $3m^2 + 56m + 36 = 0$
The tangents from the origin O to C_1 touch C_1 at the points A and B.
(b) Find the coordinates of the points A and B.
Another circle C_2 has equation $x^2 + y^2 = 13$. The tangents from the point $(4, -7)$ to C_2 touch it at the points P and Q.
(c) Find the coordinates of either the point P or the point Q.
Mark scheme on next slide.

(This is not a tangent/chord question but is worthwhile regardless!)

3 *[STEP 2005 Q6]*
(i) The point A has coordinates $(5, 16)$ and the point B has coordinates $(4, -4)$. The variable P has coordinates (x, y) and moves on a path such that $AP = 2BP$. Show that the Cartesian equation of the path of P is $(x + 7)^2 + y^2 = 100$.
(ii) The point C has coordinates $(a, 0)$ and the point D has coordinates $(b, 0)$. The variable point Q moves on a path such that $QC = k \times QD$, where $k > 1$.
Given that the path of Q is the same as the path of P, show that $\frac{a+7}{b+7} = \frac{a^2+51}{b^2+51}$
Show further that $(a + 7)(b + 7) = 100$, in the case $a \neq b$.

Mark scheme on next slide.

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Mark Scheme for Extension Question 2

32

Coordinate Geometry

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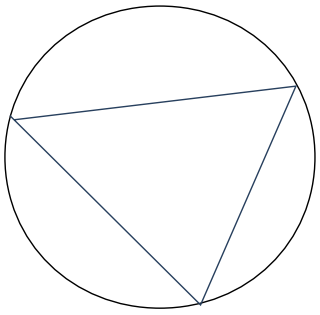
Mark Scheme for Extension Question 3

33

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Triangles in Circles

34



We'd say:

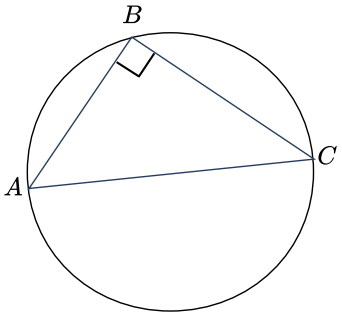
- The triangle **inscribes** the circle.
  (A shape inscribes another if it is inside and its boundaries touch but do not intersect the outer shape)
- The circle **circumscribes** the triangle.
- If the circumscribing shape is a circle, it is known as the **circumcircle** of the triangle.
- The centre of a circumcircle is known as the **circumcentre**.
-

Coordinate Geometry

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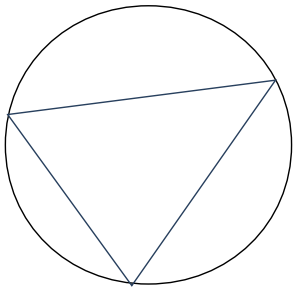
Triangles in Circles

35



If $\angle ABC = 90^\circ$ then:
- **$\mathbf{AC}$ is the diameter of the circumcircle of triangle $\mathbf{ABC}$.**
-

Similarly if AC is the diameter of a circle:
- **$\angle \mathbf{ABC} = \mathbf{90^\circ}$ therefore $\mathbf{AB}$ is perpendicular to $\mathbf{BC}$.**
  - **$\mathbf{AB^2 + BC^2 = AC^2}$**



Given three points/a triangle we can find the centre of the circumcircle by:

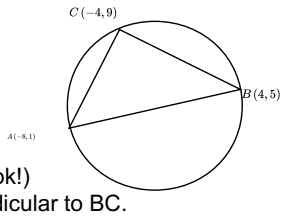
- **Finding the equation of the perpendicular bisectors of two different sides.**
- **Find the point of intersection of the two bisectors.**

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Example

36

[Textbook] The points $A\,(-8,1), B\,(4,5), C\,(-4,9)$ lie on a circle.
a) Show that AB is a diameter of the circle.



Method 1:
Show that $AC^2 + BC^2 = AB^2$

Method 2: (not in textbook!)
Show that AC is perpendicular to BC.

b) Hence find the equation of the circle.

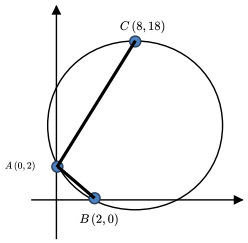
Coordinate Geometry

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Example

37

The points $A\,(0,2), B\,(2,0), C\,(8,18)$ lie on the circumference of a circle. Determine the equation of the circle.

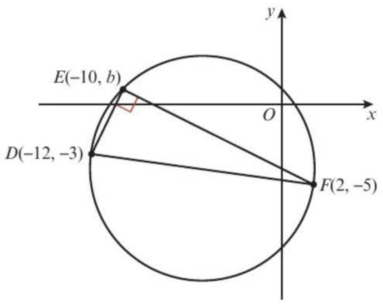


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Exercise 6F - Y1 Page 131-132

38

8 The points $D(-12,-3), E(-10,b)$ and $F(2,-5)$ lie on the circle C as shown in the diagram.
Given that $\angle DEF = 90^\circ$ and $b > 0$
a show that $b = 1$ \hfill(5 marks)
b find an equation for C. \hfill(4 marks)



a $D(-12,-3), E(-10,b), F(2,-5)$

Coordinate Geometry

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Exercise 6F - Y1 Page 131-132

39

9 A circle has equation $x^2 + 2x + y^2 - 24y - 24 = 0$.
a Find the centre and radius of the circle.
b The points $A(-13, 17)$ and $B(11, 7)$ both lie on the circumference of the circle. Show that AB is a diameter of the circle.
c The point C lies on the negative x-axis and the angle $ACB = 90°$. Find the coordinates of C.

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Exercise 6F - Y1 Page 131-132

40

Extension:

1 [STEP 2009 Q8 Edited] If equation of the circle C is $(x - 2t)^2 + (y - t)^2 = t^2$, where t is a positive number, it can be shown that C touches the line $y = 0$ as well as the line $3y = 4x$.

Find the equation of the incircle of the triangle formed by the lines $y = 0, 3y = 4x$ and $4y + 3x = 15$.
Note: The incircle of a triangle is the circle, lying totally inside the triangle, that touches all three sides.